Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/02/2004
PI Name:	Ezekoye, Ofodike Ph.D.		
Project Title:	Resonant Acoustic Control of Turbulent Diffusion Flames for Pollutant Reduction		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dezekoye@mail.utexas.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	512-471-3085
Organization Name:	University of Texas - Austin		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	ETC 5 160		
PI Web Page:			
City:	Austin	**State:**	TX
Zip Code:	78712-1063	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hegde, Uday	**Contact Phone:**	216-433-8744
Contact Email:	Uday.Hegde@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clemens, Noel T (University of Texas at Austin)		
Grant/Contract No.:	NNC04AA04A		
Performance Goal No.:			
Performance Goal Text:			
	A technology roadmap developed in 1999 by the industrial combustion community under funding from the Department of Energy identified three core R&D needs for industrial combustion over the next 20 years. The highest priority needs were identified to be burner technologies, boiler systems and process heating systems. These technologies impact a wide range of industrial applications, such as power generation, heat treatment, kilns, drying, copper smelting and glass melting. Burners represent a cross-cutting technology and improvements in burner technology will have major payoffs. The industrial combustion roadmap identifies constraints and requirements for next generation burners that include emissions constraints, stabilization issues, and economical premixing mechanisms. The roadmap further states that the advanced burner of the future “would also be able to change the flame pattern or heat delivery from the burner” and that “one concept of a ‘smart’ burner incorporates non-steady-state combustion where fluctuations are controlled rather than random.” There is a clear need to scientifically explore all innovative burner concepts since it is not obvious what		

Task Description:	directions in burner designs will prove fruitful in the future. We are investigating an essentially unexploited burner strategy that uses high-intensity, high-frequency forcing of a fuel jet to achieve substantial partial-premixing of initially nonpremixed reactants. Our results show that it is possible to produce dramatic changes in the structure and luminosity of a turbulent nonpremixed jet flame with the application of strong forcing of the jet flow. For example, a turbulent nonpremixed methane jet flame, which exhibits substantial yellow luminosity due to emission from soot, can be made to turn completely blue and to shorten to about 1/3 of its original length, by the application of the forcing. These intense pulsations are achieved by modulating the fuel flow with a high-speed valve at a frequency that coincides with the organ-pipe resonance frequency of the fuel tube. The combination of rapid fuel flow rate modulation, coupled with the organ pipe resonance of the fuel tube results in very large amplitude, high –frequency modulation of the fuel gas pressure and velocity. The resulting flow exhibits a high degree of partial premixing of the fuel and air. Using direct flame photography, hotwire velocity measurements, planar laser induced fluorescence, and planar laser mie scattering we find that the changes in flame properties are due to ingestion (sucking) of ambient air into the fuel tube, vortex-ring entrainment, and turbulence generated by the unsteadiness. This form of high- amplitude, high-frequency forcing of the fuel jet differs significantly from most other forcing schemes, which tend to use low-amplitude, high-frequency (e.g. with a speaker) or high-amplitude, low-frequency (e.g. flame puffs) forcing. In this project we will study and detail the mechanisms responsible for significant partial premixing in resonantly driven, transitional and turbulent jet diffusion flames. The ability to predict the flame length, and lifted flame length for these pulsed flames, and also the ability to discuss transitions between the luminous and nonluminous flame conditions requires careful experimental work. These experiments are to be divided between low gravity and normal gravity conditions. Differences in flame properties between the low gravity conditions and normal gravity conditions will help us to decouple buoyancy-induced modifications to mixing and flame length from acoustically-induced modifications. Some normal gravity experiments will use conventional measurements of gas species simply to establish the basic characteristics of this type of flame. Numerical modeling will be used to better understand the experimental results. The objectives of this research are to: 1) characterize the mechanisms for increased near field mixing in strongly pulsed high frequency jet flames, 2) use the microgravity environment to develop an understanding of the role of buoyancy for near field mixed flames in modifying flame characteristics, 3) develop scaling relationships that can be used to design burners for practical applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Flame Measurements The effects of acoustic pulsations on the global parameters of methane jet diffusion flames like flame length and luminosity can be obtained from the flame measurements. To this effect flame measurements were made using a digital camera at resonant and non-resonant pulsing conditions of the fuel flow issuing out of the nozzle at different flow-rates. The luminosity and the flame length of the diffusion flames show a dramatic change as the flow is pulsed near resonant pulsing conditions, suggesting intense and localized mixing of the fuel and the surrounding air. Velocity Measurements Velocity measurements were made at the nozzle exit as well as at several downstream distances using a single-wire, constant temperature hotwire anemometer for air flow through the nozzle. Due to the use of single-wire hotwire probe, the velocity signal shows a large peak and then a small peak when pulsed at resonance frequency corresponding to the positive peak and negative peak, respectively. When pulsed at the second resonance frequency the velocities are still large compared to the un-pulsed velocity, however, the positive as well as the negative velocities are smaller than those of the first resonance frequency. For non-resonance pulsing, we found that the positive velocity increases to about twice that of the peak velocity at the un-pulsed condition. However, there is a negligibly small negative velocity which might be the reason for the non-resonantly pulsed flame looking similar to the un-pulsed flame. Power spectral desnity calculations were performed. These show that the distance where the coherent structures break down depends on the Reynolds number of the flow and the frequency of forcing. When the coherent structures break down, the spectral density shows a power law variation with a power law scaling of -5/3. Planar Laser Induced Fluorescence and Planar Laser Mie Scattering Using laser based diagnostic techniques like planar laser induced fluorescence (PLIF) and planar laser Mie scattering (PLMS), we investigated the physical mechanisms that lead to enhanced partial-premixing. Further insight into the mechanisms responsible for the effects due to pulsing of the flame was gained by imaging the light scattered from glycerol fog fed into the jet. This planar laser Mie scattering (PLMS) images were obtained at a rate of 8000 frames per second. The PLMS data showed that when the forcing frquency exceeded a critical value, the vortices generated at the nozzle exit plane were unable to separate from the nozzle. Numerical Analysis Simulations have been performed using a high-order finite difference code. These simulations have helped clarify the experimental findings. A limitation of the simulations is that only laminar flows can be simulated. Nevertheless, the laminar unsteady dynamics have clarified how the vortex formation and convection processes are modified by the unsteady flow. Both combustion and cold flow simulations have been performed to uncover the dynamics of the pulsed operation. With higher frequency forcing at the same amplitude level, we find that the vortex never escapes the nozzle region. This is similar to the experimental findings at the second resonance. If we push the amplitude even larger at the frequencies where the vortex does not initially escape for low amplitude forcing, we are able to dislodge the vortex.
Bibliography Type:	Description: (Last Updated: 06/02/2008)
Papers from Meeting Proceedings	Lakshminarasimhan, K., Whitaker, J., Clemens, N.T. , and O. A. Ezekoye, "Diffusion Flame Properties in Transitional Pulsed Jet Flows" Central States Section Combustion Institute Meeting, Austin, TX Mar-2004
Papers from Meeting Proceedings	Barve, V. V and Ezekoye, O. A. "Computations of Time-Varying Flame Properties for Laminar, Methane-Air Diffusion Flames" Central States Section Combustion Institute Meeting, Austin, TX Mar-2004